

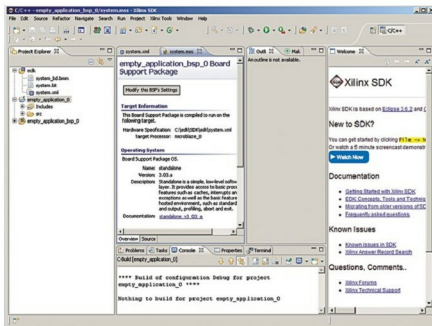


Fig. 18: Xilinx SDK window

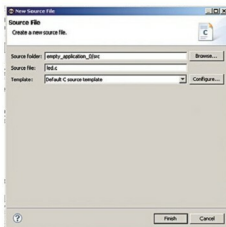


Fig. 19: New Source File window

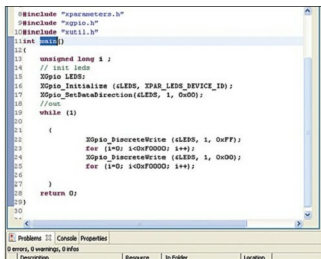


Fig. 20: Program compilation

the file, it is automatically compiled. You should get zero errors, as shown in Fig. 20

The C code first initialises the LEDs port and then sets the port as output port. Then the LEDs are switched on and off using a delay created by the 'for' loop

```
#include "xparameters.h"
#include "xgpio.h"
#include "xutil.h"

//=====Program=====//
int main () {
    unsigned long i;
    //Init LEDs
    XGpio LEDS;
    XGpio_Initialize(&LEDS, XPAR_LEDS_DEVICE_ID);
    XGpio_SetDataDirection(&LEDS, 1, 0x00);
    //Out
    while (1)
    {
        XGpio_DiscreteWrite(&LEDS, 1, 0xFF);
        for (i=0; i<0xF0000; i++);
        XGpio_DiscreteWrite(&LEDS, 1, 0x00);
        for (i=0; i<0xF0000; i++);
    }
    return 0;
}
```

Select Xilinx Tools → Program FPGA. The bitstream gets loaded to the FPGA. The LEDs start blinking. If they do not blink, press the reset switch used in the design. On pressing the reset switch they will start blinking.

Conclusion

With this part, the series of articles on Designing with FPGAs is concluded. The articles were written with an aim to provide the electronics engineers, professionals and hobbyists an insight into different aspects of system design using FPGAs. ●

Concluded

Varsha Agrawal is a scientist at Laser Science and Technology Center (LASTEC), a premier DRDO lab working in the field of laser-based defence systems. She has more than 13 years of R&D experience in the design and development of a variety of systems for defence-related applications. She has authored two books and published more than 20 research papers and technical articles

Click Finish

24. The wizard that follows allows you to create a template software application for the project. The default is 'Hello World' example. We want to create a blank application. Select Empty Application and click Next, as shown in Fig. 17. Click Finish

25. An SDK window appears as shown in Fig. 18

26. In the Project Explorer, open the tree empty_application_0 → src. There is no source file in the folder. We need to add the source file. Select src folder and go to File → New → Source File. In the

dialogue box that opens, specify the name of the source file to be created and the folder in which to put it, as shown in Fig. 19. Click Finish. If you now expand the src folder, you will find led.c inside that folder

27. Click on led.c and paste the code given below onto it. Save the file. When you save